

Choosing a 2.5D chip for your application

According to Ganesan, the circuit designer must give paramount importance to the total cost when choosing 2.5D/3D chips. Total cost includes TSV cost, packaging cost, die cost, test cost and yield cost.

Cost, if not evaluated properly, could be a bigger headache than performance when marketing the end product. Incorrect usage or assumptions about 2.5D/3D design and manufacturing could lead to a solution that is overpriced for its target market.

Design considerations

The design considerations for using 2.5/3D chips depend largely on the application.

"For a design using stacked dynamic random-access memory (DRAM), the key is to determine how to adapt to the increased bandwidth capability. In most cases, it is not currently possible to split logic functionality between vertically adjacent chips, but it can be split horizontally using an interposer," explains Ganesan. Peckham says, "In most cases, a 2.5D device looks and behaves just like a monolithic device. The interconnect between dies is slightly longer than the interconnect delay within a die, so a field-programmable gate array (FPGA) designer may think about floor-planning his design accordingly. But the design tools take this into account, so that the designer can use design constraints and simulations in the usual way."

Two major constraints while using 3D technology are heat and cost. It is important to consider the thermal aspects of any 2.5D/3D system and to ensure that no chip is placed in an environment that is too hot for it, and that heat is effectively dissipated from all parts of the design. Similarly, a bad design can lead to an overpriced system that nobody wants!

What design tools to use

Major electronic design automation (EDA) companies have tool chains for the true 3D development, but these are

still in very early stages. Chip designers should count on developing very tight relationships with their EDA, intellectual property (IP) and silicon vendors in order to ensure that their designs work.

Synopsis has a complete tool-chain for 2.5D/3D design, which includes interconnect simulation, test automation, memory test and repair IP, physical implementation, custom implementation, physical verification, sign-off and circuit simulation. Similarly, Calypto, Cadence and Mentor Graphics also have design tools for 2.5D/3D chips.

Chip makers are also coming up with their own design tools to ensure that designers are able to make maximum use of the extended capabilities of 2.5D/3D devices. Xilinx, which offers the Virtex-7 range of 2.5D devices, has built a design suite called Vivado to enable the circuit designers to use full power of its 2.5D ICs.

Experts feel that the 3D paradigm will not necessarily require new tools but just the imbibing of 3D awareness into existing design tools for better floor-planning, place and route, extraction, analysis and IC/package co-design.

High-level planning and co-design

High-level planning and co-design are very important to reduce power dissipation, maximise performance and bring down the cost. Proper path finding tools must be used for high-level planning and partitioning. The first task is to partition system functions into hardware and software. Then designers will have to consider which proven die-level IP they should use for various functions, what process technologies (Si, Ge, GaAs, etc) need to be employed, which feature size should be used, how to place the components and interconnects, whether an interposer is needed, how to use it, and so on.

Next, engineers should attempt to co-optimize die and package by building a top-level netlist, optimising input/output (I/O) connections and power supply lines, creating a pack-

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age layout in conjunction with floor-planning and routing on the individual die, and predicting and avoiding thermal power, signal integrity and noise problems.

Coming to the board design, the designers should consider where the 3D IC package will be placed, how well it can be cooled and how the interconnects will be routed. Co-design of IC and printed circuit board (PCB) can optimise the placement and rotation of components and reduce the number of layers in the board. In any case, the circuit designers must remain prepared to face significant challenges when using 2.5D/3D chips in any project, as is the case with such nascent technologies.

Remember—its not just a chip, but a whole system

The success of the 3D technology depends on the proper understanding of the concept and a holistic paradigm shift—engineers need to understand that at the 3D level they will not simply be designing chips or circuits at standalone levels but will be creating a whole system on a single multi-die chip. These systems will integrate digital, analogue and radio frequency (RF), logic, memory functions, micro-electrical mechanical systems (MEMS), multi-core or many-core processor architectures, and sophisticated software.

Ganesan signs off saying, "2.5D is likely to be the choice for the next few years as 3D technology ramps up, but eventually true 3D will become dominant." ●

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